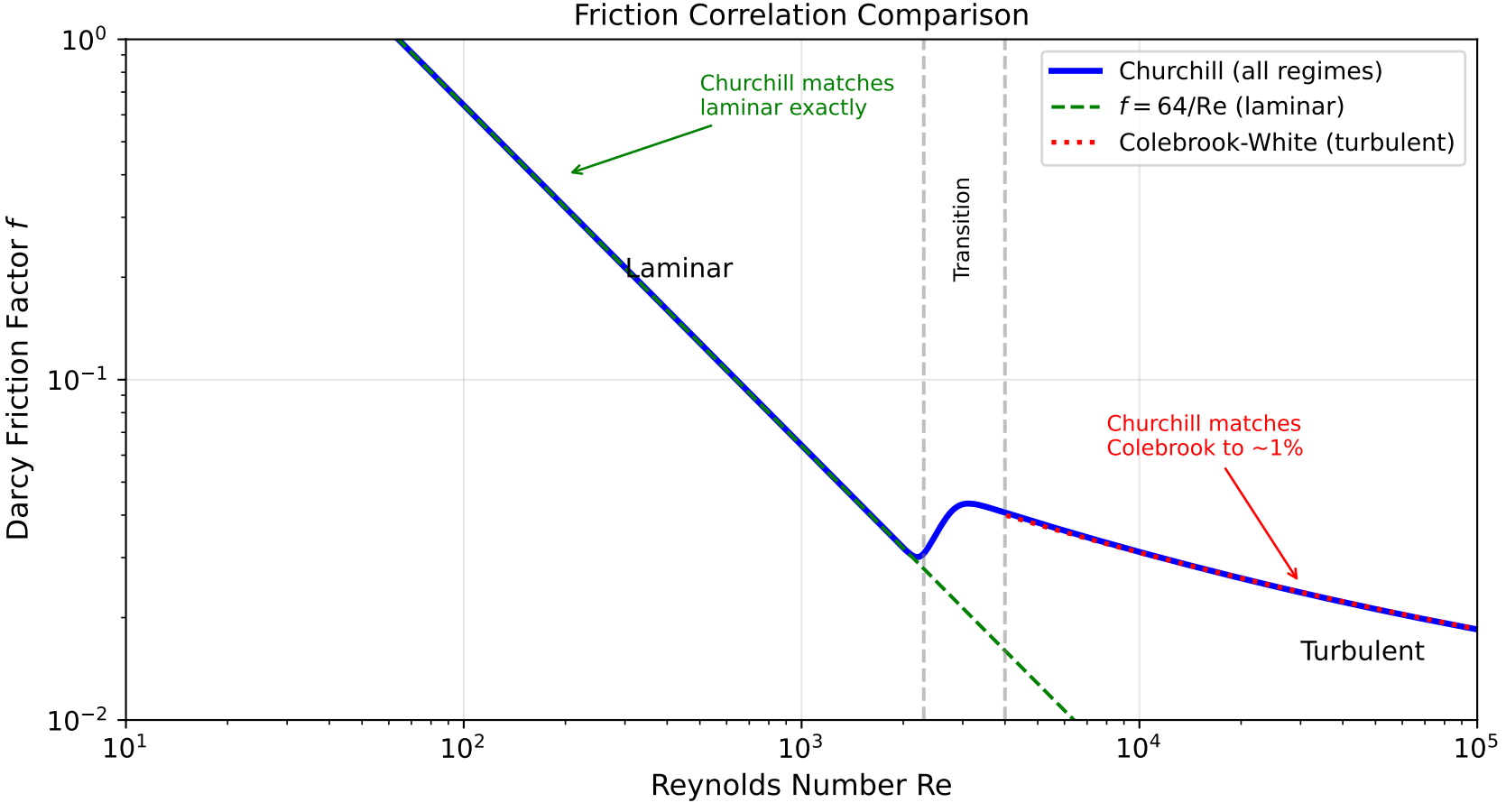
Friction Correlation Comparison
Churchill (all regimes)
$f = 64/\mathrm{Re}$ (laminar)
Colebrook-White (turbulent)
Churchill matches
laminar exactly
Laminar
Transition
Churchill matches
Colebrook to ~1%
Turbulent
Darcy Friction Factor f
Reynolds Number Re
10^{0}
10^{-1}
10^{-2}
10^{1}
10^{2}
10^{3}
10^{4}
10^{5}